

EVEN SUBDIVISIONS ARE CONTRACTIBLE

ABSTRACT. We prove that a bipartite graph is contractible in the sense of graph schemes whenever every vertex in one part has degree at most two. Consequently, replacing every edge of a loopless multigraph by a path of positive even length produces a contractible graph. The proof projects a coloured scheme onto a family of graphic matroids. Each edge label occurs in at most two projections, and a component count verifies the matroid union inequalities needed for simultaneous disjoint spanning trees.

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1. STATEMENT

All graphs are finite. Target and scheme graphs are simple; auxiliary multigraphs may have parallel edges but no loops. Following Kündgen, Pelsmajer and Ramamurthi [2, Definitions 1.1 and 2.1], an H -scheme in a graph G with $V(H) \subseteq V(G)$ is a family $(P_{uv} : uv \in E(H))$ of paths satisfying two conditions:

- (1) P_{uv} joins u to v and contains no other vertex of H ;
- (2) whenever a nonempty collection of scheme paths has a common vertex, their corresponding edges in H have a common endpoint.

A *rooted H minor* in G is a family $(C_v : v \in V(H))$ of pairwise disjoint connected vertex sets, with $v \in C_v$ for every v , such that C_u and C_v are adjacent whenever $uv \in E(H)$. The graph H is *contractible* if every graph containing an H -scheme contains a rooted H minor.

Theorem 1.1. *Let H be bipartite with a specified bipartition (A, B) . If $d_H(b) \leq 2$ for every $b \in B$, then H is contractible.*

Corollary 1.2. *Let F be a loopless multigraph. Replace each edge of F by a path of positive even length, using mutually disjoint new internal vertices. The resulting simple graph is contractible.*

Proof. Place every original vertex, and every even-position internal vertex of each replacement path, in A . Place the odd-position internal vertices in B . Every vertex of B has degree two, so Theorem 1.1 applies. Isolated vertices of F may be retained. $\square$

Theorem 1.1 includes $K_{2,n}$ for every n , with the part of order n as B . Corollary 1.2 includes the graph obtained by subdividing every edge of any

simple graph exactly once. Throughout, “even” refers to the length of each replacement path.

2. PACKING SPANNING TREES WITH SHARED LABELS

Lemma 2.1. *Let J be a finite index set. For each $j \in J$, let M_j be a connected multigraph with a distinguished vertex r_j and an edge set labelled bijectively by a finite set E_j . Suppose that M_j is the union of a family $\mathcal{Q}_j$ of paths whose edge sets partition E_j , all these paths contain r_j , and every other vertex belongs to at least two of them. Suppose also that each element of $E = \bigcup_j E_j$ belongs to at most two sets E_j . Then there are pairwise disjoint sets $T_j \subseteq E_j$ such that T_j is a spanning tree of M_j . Trivial paths are allowed; an empty family is allowed when $V(M_j) = \{r_j\}$.*

Proof. Fix $X \subseteq E$. Let $c_j(X)$ count the components, including isolated vertices, of the spanning subgraph of M_j with edge labels $E_j \cap X$. Put $m_j = |\mathcal{Q}_j|$. The component containing r_j meets all m_j paths, and every other component meets at least two. Conversely, a path Q meets at most $1 + |E(Q) \setminus X|$ components: deleting those edges leaves that many path pieces, each contained in one component. Thus

$$m_j + 2(c_j(X) - 1) \leq \sum_C |\{Q \in \mathcal{Q}_j : V(Q) \cap V(C) \neq \emptyset\}| \leq m_j + |E_j \setminus X|.$$

The last bound uses the partition of E_j into the path edge sets. It follows that

$$(1) \quad c_j(X) - 1 \leq \frac{1}{2}|E_j \setminus X|.$$

Extend the graphic matroid of M_j to the common ground set E by declaring all labels outside E_j to be matroid loops. Its rank function ρ_j satisfies

$$\rho_j(E) = |V(M_j)| - 1, \quad \rho_j(X) = |V(M_j)| - c_j(X).$$

Summing (1) and using the label multiplicity bound gives

$$(2) \quad \sum_j (\rho_j(E) - \rho_j(X)) \leq \frac{1}{2} \sum_j |E_j \setminus X| \leq |E \setminus X|.$$

The finite matroid union rank formula [1] states that

$$\rho_{\bigvee_j \mathcal{M}_j}(E) = \min_{X \subseteq E} \left(|E \setminus X| + \sum_j \rho_j(X) \right).$$

Equation (2), together with $X = E$, makes this rank equal to $\sum_j \rho_j(E)$. Write a maximum independent set of the union as $\bigcup_j I_j$, where each I_j is independent in $\mathcal{M}_j$. Equality throughout

$$|\bigcup_j I_j| \leq \sum_j |I_j| \leq \sum_j \rho_j(E)$$

forces the I_j to be pairwise disjoint bases. A base contains no matroid loops. Therefore $T_j = I_j \subseteq E_j$ is the required spanning tree. $\square$

3. PROOF OF THEOREM 1.1

By the coloured-scheme reduction [2, Lemma 3.3], an H -scheme has a rooted minor whose underlying graph G supports a *coloured H -scheme*. This consists of a proper colouring $f : V(G) \rightarrow V(H)$ fixing each root, such that P_{uv} uses only colours u, v and every nonroot has degree at least four. We remove isolated roots before this reduction and restore them as singleton branch sets at the end. The graph with no remaining roots is immediate.

The paths alternate colours, every edge belongs to exactly one scheme path, and each nonroot lies on at least two scheme paths [2, Remark 3.2(1), (2), (6), (7)]. Set

$$A_a = f^{-1}(a) \quad (a \in A), \quad L_b = f^{-1}(b) \quad (b \in B).$$

If $d_H(b) \leq 1$, then $L_b = \{b\}$, since no nonroot of colour b can lie on two distinct paths incident with b . If $d_H(b) = 2$, every nonroot of colour b lies internally on both paths incident with b . Put

$$E = \bigcup_{b \in B} (L_b \setminus \{b\}), \quad E_a = \bigcup_{b \in N_H(a)} (L_b \setminus \{b\}).$$

These labels are actual vertices of G . Each label belongs to exactly two sets E_a , one for each neighbour of its degree-two root colour.

Construct M_a on vertex set A_a as follows. For each $b \in N_H(a)$ and $x \in L_b \setminus \{b\}$, put an auxiliary edge labelled x between the two neighbours of x on P_{ab} . Both neighbours have colour a , and they are distinct because P_{ab} is a path. Different labels may give parallel edges. Suppress all such vertices x on P_{ab} and remove its terminal vertex b . This leaves a path Q_b^a in M_a from a to the A_a -neighbour of b ; it is trivial if $P_{ab} = ab$. Its edge-label set is exactly $L_b \setminus \{b\}$.

As b ranges over $N_H(a)$, the paths Q_b^a partition E_a , cover A_a , and all contain a . In particular, M_a is connected. Every vertex of $A_a \setminus \{a\}$ lies internally on at least two distinct scheme paths incident with a , so it remains on at least two projected paths. Hence the family $(M_a : a \in A)$ satisfies Lemma 2.1. Choose its pairwise disjoint spanning-tree label sets T_a .

In the reduced graph G , define

$$C_a = A_a \cup T_a \quad (a \in A), \quad C_b = \{b\} \quad (b \in B).$$

These sets are pairwise disjoint: the A_a are different colour classes, the T_a are disjoint sets of nonroot B -colour vertices, and no T_a contains a root. Each tree edge labelled x lifts to the two-edge path through x in G , so $G[C_a]$ is connected. Moreover $a \in C_a$. For each $ab \in E(H)$, the last edge of P_{ab} joins the root b to a vertex of $A_a \subseteq C_a$. Thus these branch sets form a rooted H -minor model.

Finally compose this model with the rooted minor reduction. Replace every vertex in each branch set by its connected preimage in the original graph. Disjointness, connectivity, root containment and required adjacencies are preserved. Restoring the isolated roots completes the proof. The

singleton assertion for the B branch sets applies only in the reduced coloured scheme. $\square$

4. SCOPE

The contractible targets supplied by Corollary 1.2 have unbounded treewidth, since they include a once-subdivision of K_t for every t . This does not assert that contractibility is minor-closed or that K_t itself is contractible. Theorem 1.1 also covers theta graphs whose three paths all have even length; it does not cover the case of three odd-length paths. It therefore addresses only part of the bipartite-theta question in [2, Section 8, Question 3]. Neither all bipartite graphs nor $K_{3,3}$ are covered. The theorem makes no conclusion about Hadwiger’s conjecture or its K_7^- analogue.

REFERENCES

- [1] J. Edmonds, *Submodular functions, matroids, and certain polyhedra*, in Combinatorial Structures and Their Applications, Gordon and Breach, 1970, 69–87.
- [2] A. Kündgen, M. J. Pelsmajer and R. Ramamurthi, *Finding minors in graphs with a given path structure*, J. Graph Theory **79** (2015), 30–47. doi:10.1002/jgt.21812. Primary preprint: arXiv:1207.6141.